



partageons les connaissances au profit des communautés rurales  
sharing knowledge, improving rural livelihoods

## **Rural Radio Resource Pack**

**No 06/3**

# **AVIAN FLU**



The Technical Centre for Agricultural and Rural Cooperation (CTA) was established in 1983 under the Lomé Convention between the ACP (African, Caribbean and Pacific) Group of States and the European Union Member States. Since 2000, it has operated within the framework of the ACP-EC Cotonou Agreement.

CTA's tasks are to develop and provide services that improve access to information for agricultural and rural development, and to strengthen the capacity of ACP countries to produce, acquire, exchange and utilise information in this area.

#### *Rural radio*

Radio remains, despite all the interest in the new ICTs, one of the most important communication tools in ACP rural communities. CTA began supporting rural radio back in 1991. Every year since then we've produced a set of Rural Radio Resource Packs (RRRPs).

Each pack is on a specific topic – anything from crop storage and cassava to small ruminants and soil fertility. The choice of topics depends on what ACP partners suggest. The number of topics covered has now reached 51. Inside each pack are materials for a radio programme on that topic – interviews on cassette or CD, a transcription and a suggested introduction for each interview, technical information on the topic, advice for how the pack can be used and a questionnaire for users to provide feedback to CTA.

You can find most of the RRRP material on Anancy, CTA's virtual library (<http://www.anancy.net>).

CTA  
Postbus 380  
6700 AJ Wageningen  
The Netherlands

Website: [www.cta.int](http://www.cta.int)



partageons les connaissances au profit des communautés rurales  
sharing knowledge, improving rural livelihoods

## **Rural Radio Resource Pack**

**No 06/3**

# **AVIAN FLU**

CTA Technical Centre for Agricultural and Rural Co-operation  
Postal Address: Postbus 380, 6700 A J Wageningen,  
The Netherlands  
Telephone (31) (0) 317 467100 Fax (31) (0) 317 460067  
<http://www.cta.int>

produced for CTA by  
WRENmedia, Fressingfield, Eye, Suffolk, IP21 5SA, UK  
<http://www.wrenmedia.co.uk>



# CTA

## *Rural Radio Resource Pack - 06/3*

### **Avian Flu**

#### **TECHNICAL INFORMATION**

##### **Introduction**

Since 2003, the H5N1 avian flu virus (also known as Highly Pathogenic Avian Influenza – HPAI) has spread from Southeast Asia to more than 30 countries in the Middle East, Western Asia, Eastern Europe, the Indian subcontinent and sub-Saharan Africa. The crisis has caused billions of dollars of losses; more than 200 million birds have been culled or have died from the disease. Over 200 people have been infected by the disease, and more than 100 have died, mostly children and young adults. Fears about the risk to human health have also led to lower prices for poultry and poultry products.

In Africa, most outbreaks of avian flu have occurred in Nigeria (69 outbreaks), Egypt (40 outbreaks), and Sudan (15 outbreaks)\*, with smaller numbers of outbreaks having occurred in Burkina Faso, Cote D'Ivoire, Niger, Cameroon and Djibouti. It is predicted that the spread of avian flu will reduce poultry production in Africa and could also reduce consumption of poultry products by five per cent. This would have a major effect both on farmers' livelihoods and on food security; in Africa, poultry contributes nearly 30 per cent of total meat consumed.

Backyard poultry are vital in Africa, not only as a source of food, but as a form of saving, as a source of income, for barter and for social exchange. A widespread outbreak of avian flu would not only reduce the availability of food but would have serious impacts on livelihoods. Controlling any outbreaks that occur is therefore a major priority for African governments and deserves to be a high priority for the international organisations and donors that support them.

##### **Control strategies**

To control avian flu, it must be detected rapidly when an outbreak occurs, and it must be quickly and effectively eradicated in order to prevent economic losses and limit infections to humans. The most important control strategies are:

- Disease surveillance, so incidents are detected and responded to quickly
- Improving biosecurity on poultry farms
- Controlling the movement of birds and poultry products in outbreak areas
- Temporary closure of markets and control of borders in outbreak areas
- Culling (humane slaughter) of infected poultry (this includes sick and healthy birds in affected flocks)
- Communication and education of public

Having a well-informed public is extremely important in controlling outbreaks. People must be aware of the risks to poultry and humans, of disease symptoms in birds, and the danger of allowing domestic poultry to mix with wild birds, which may be carrying the virus without showing any physical symptoms.

\* data from OIE – World Organisation for Animal Health, figures up to 27<sup>th</sup> October 2006

This Rural Radio Resource Pack can help you to raise awareness among rural radio audiences about the risks of avian flu, and information they need to know in order to ensure that if an outbreak occurs, it can be quickly detected and controlled.

## **Topics covered in this pack**

### **Recognising symptoms of avian flu in domestic poultry**

Wild birds infected with H5N1 avian flu may not show any physical symptoms of the disease, but can still spread it to domestic poultry. Farmers should therefore avoid all contact between their birds, and wild birds, even if those wild birds appear to be healthy.

Signs of avian flu in domestic poultry are similar to some other poultry diseases, especially Newcastle disease. These signs include:

- general appearance of ill-health, e.g. lack of energy and activity
- massive reduction in egg laying by layers
- heads swell and bare parts of the bird (e.g. combs and feet) become reddened

However, avian flu can spread very quickly within a flock, and can cause sudden death in 100 per cent of the flock almost without warning. With free range birds, farmers may fail to notice such symptoms, but will simply find that birds have suddenly died. All these symptoms are described in the first interview, ***Recognising and responding to disease symptoms***.

This interview also includes mention of what farmers should do if they see these symptoms in their birds. Avian flu looks like some other diseases, such as Newcastle disease. Farmers should report any suspicious illness or deaths in their birds immediately to either veterinary or human health centres. Only through laboratory tests will a definite diagnosis of avian flu be possible.

### **Protecting backyard/village poultry from infection**

#### **Enclosures**

If birds are allowed to scavenge freely for food, and are drinking and washing in open water such as ponds and streams, it will be very difficult to protect them from an outbreak of avian flu. For this reason, one of the most important steps that poultry keepers can take to protect their birds, is to keep them in some kind of enclosure. Some enclosures are made of wire mesh; these are usually known as 'fowl runs' or 'chicken runs'. But birds can be kept in a enclosure made from locally available materials such as grass, mud bricks or metal sheets.

One important point about an enclosure is that it should be bird and rat proof. This means that the poultry within the enclosure, and their feed and water, are protected from contact with both wild birds and rodents, which could otherwise bring disease into the flock from outside. Farmers who cannot enclose their birds should at least be careful not to leave piles of poultry food on the ground, since these can attract wild birds.

#### **Feed and water**

If farmers are giving feed to their birds, they must ensure that the feed is free from disease. They can do this by buying feed from a recognised, regulated source or manufacturer. If farmers can not afford to do this, they should at least ensure that the feed they are using is stored safely where it can not be contaminated by wild birds or rodents.

Ensuring clean water supply is another priority. Farmers should not take water from streams and ponds where wild birds such as ducks drink, swim and bathe. If farmers are uncertain about the quality of their water, they may decide to boil it before they give it to their birds.

This would be especially important for farmers in areas visited by migratory birds where risk of infection is high.

### **Introducing new birds**

Farmers should not mix birds from different sources. The safest way of rearing poultry is called the 'all in, all out' system. This means that farmers buy a batch of chicks of the same age from the same hatchery, rear them to a saleable weight, and then sell the entire batch at one time. The chicks are kept separately as a batch and are not mixed with any other birds during this period. Once the batch has been sold, the poultry house is thoroughly cleaned and disinfected, so that no diseases can be passed on to the next batch. This model is widely used by commercial farmers, but is not normally practised by small-scale poultry keepers.

If farmers want to introduce new birds to their flock, they should keep the new bird or birds separate from the flock for at least two weeks before they allow it to have contact with the other birds. During this two week period, the farmer should watch closely for any signs of disease in the new birds; obviously if they develop any disease symptoms they should not be brought into the flock.

### **Don't mix different species**

It is also important that poultry keepers do not mix different types of poultry (e.g. chickens, ducks, geese, turkeys, guinea fowls, pigeons) in the same enclosure or area. One reason for this is that some poultry, such as ducks, may become infected with a disease but not immediately show symptoms. Such birds could then spread the disease to the whole flock.

The interview ***Protecting village and backyard poultry*** begins by looking at some of the major routes by which avian flu can be spread to backyard poultry, and then discusses what village poultry keepers can do to protect their birds from the disease. This is followed by ***The importance of enclosures***, in which a backyard poultry keeper from an urban area describes how she manages her birds in a fowl run, and a veterinary officer gives other practical tips on enclosing village poultry.

### **Protecting commercial poultry farms from infection**

Many of the same rules that apply to backyard poultry also apply to commercial farms: using clean sources of food and water, buying new birds from sources that are registered and disease-free, practising an 'all in, all out' system of rearing. When using the 'all in, all out' system, a rest period of one week between flocks is strongly advised, and careful cleaning of the poultry houses is essential.

Commercial farms should also have tight biosecurity rules. For example:

- controlling who and what enters the farm;
- ensuring that anyone entering the farm is disinfected and their boots or shoes are cleaned with disinfectant;
- keeping poultry houses and cages very clean;
- daily cleaning of all equipment used in the poultry houses, such as feeding scoops, shovels, brooms and feeding trays;
- not sharing equipment between farms;
- disinfecting any equipment that has been used on another farm

People who work on the poultry farm will also have to obey tight hygiene rules, for example having a uniform or overall which they only wear on the farm, to ensure that no diseases are taken off the farm, or brought on to it. Farm workers must be prohibited from keeping poultry in their homes, or else they risk transferring disease from their own birds to the commercial flock. If any bird on the farm becomes sick, it should be separated from the other birds

immediately. The authorities should be informed so that a diagnosis can be made.

These issues of what commercial poultry farmers need to do to protect their birds are discussed in the interview ***Protecting commercial poultry farms.***

### **Protecting human health**

#### **Who is at risk?**

The greatest risk to humans of catching avian flu does not come from eating diseased birds or their eggs. The greatest risk is to those who work with poultry, who handle them, feed them, collect eggs or do medical checks. The virus can be passed from infected birds to humans in several ways: in birds' saliva, in secretions from the bird's nose, in dust on feathers, in the blood and intestines of a dead bird. Those who handle, pluck or butcher infected birds are at especially high risk, as are those who keep poultry in their own home.

#### **Basic hygiene**

There are simple things that people should do to protect themselves from avian flu. In areas where no outbreak has yet occurred, poultry keepers should still have good hygiene standards. For example:

- washing hands in soap and water after handling birds;
- cleaning shoes of any poultry faeces when leaving the farm/before entering the home;
- covering hands with gloves or plastic bags when handling birds that are found dead

If farmers are suspicious of the cause of death, the dead bird should be kept in a sealed plastic bag and the veterinary authorities invited to find out the cause of death.

#### **Children**

Children may be interested in sick or dead birds. They should therefore be told not to play with or handle any sick or dead birds that they find. Instead they should tell an adult if they find a sick or dead bird. Children should be made to wash their hands after handling or feeding poultry.

#### **Proper cooking**

In areas where no outbreaks of avian flu have occurred, people should be encouraged to continue eating poultry and poultry products. However, both poultry and eggs must be cooked properly before being eaten. This will protect people from contracting a number of diseases that may be present in poultry, including salmonella and avian flu. If meat and eggs are properly cooked any viruses will be destroyed and the food will be safe to eat.

#### **When outbreaks occur**

In areas where an outbreak has occurred, people must be much more careful about contact with poultry. In such areas, all people who have contact with poultry should wear protective clothing, including a mask - to prevent them breathing contaminated dust - and gloves. People should dip their shoes in disinfectant before entering, and after leaving, poultry houses. In these areas, all poultry will be put under quarantine, so that they cannot be moved, sold or eaten. Normally, if an outbreak occurs, the authorities will slaughter all the birds in the area (up to 3-5 kilometres from the outbreak site), and dispose of the carcasses so that they cannot be eaten.

If a farmer notices that one bird in the flock is sick or has died, the farmer should assume that all the birds in the flock are infected. This means that no bird or egg from that flock should be eaten until the authorities have confirmed whether avian flu is present or not. Often when one or two birds become sick, farmers are tempted to sell the other birds quickly before they also develop disease symptoms. However, this practice is extremely dangerous, since it can

quickly spread the disease from one flock to a whole market, and from there it can be taken to hundreds of farms and homes in a very short period.

**Preventing transmission to humans** covers many of these points, and also includes a description of what symptoms occur in humans who catch avian flu, and what people should do if they think that they or someone else may have contracted the disease. One important point not contained in the interview is that if people think that someone has contracted avian flu, they should care for that person and call for a doctor immediately, but should have minimal physical contact with them, and should not sleep in the same room with them.

**Poultry meat and eggs – safe to eat?** is a short interview discussing when it is safe and not safe to eat poultry products, and gives advice for keeping children away from sick or dead birds.

**Reducing risks for poultry and people** picks up on a range of important points, including the importance of farm hygiene, regular observation of bird health, reducing risk of disease spread and transfer to humans, and control of bird movements in outbreak areas. Similar issues are tackled in **Increasing biosecurity when outbreaks occur**, which also describes how farmers should isolate any sick birds, and should cover their hands when handling sick or dead birds. The commercial farmers, the interview also includes steps to increase biosecurity, for example through the use of wheel baths for cleaning vehicle tyres.

### **Culling, movement restrictions and compensation**

When outbreaks of avian flu occur, culling (i.e. humane slaughter) of all poultry within a certain distance of the outbreak, and movement restrictions for birds, people and vehicles around the affected area are essential for controlling the disease. The authorities will set up zones around the outbreak. The area within 5 km of the outbreak - the exact distance varies in different countries - is called the **infected zone**. All domestic poultry, diseased and healthy, that live within this zone are treated as infected, and are slaughtered. Their carcasses are disposed of either by burning or by deep burial.

The infected zone is surrounded by a larger area, called the **control zone or restricted zone**. This area will have constant surveillance (monitoring), with veterinary officers inspecting farms and villages for signs of the disease. Within this area, movement of people, vehicles and birds will be very tightly controlled. All people and vehicles will have to be disinfected when passing between the infected zone and the control zone. No movement of birds between the infected zone and the control zone is allowed.

A third zone refers to the rest of the country which falls outside the infected and control zones. It is called the **disease-free zone**. Movement of people, vehicles and poultry between the control zone and the disease-free zone will be closely monitored.

### **Compensation**

When an outbreak of avian flu occurs in a flock, all birds in that flock, whether they appear sick or healthy, must be destroyed. This is the task of the authorities, not of farmers themselves. When an outbreak occurred in Cameroon in March 2006, poultry keepers within the infected zone were told to report to veterinary offices, to declare the number of poultry that they owned. Veterinary staff then visited the farms and villages to check the number of birds. These were slaughtered, and farmers were given financial compensation from the government.

Giving compensation for birds killed in culling programmes is very important. If compensation is not offered, farmers will be reluctant to report sick or dead birds, and may try to sell their poultry, thereby spreading the disease further. How much compensation should be paid for each bird is an important question. The amount of money needs to be sufficient to

encourage farmers to report any diseased birds. However, if the amount offered is too much, farmers have no incentive to try to protect their flocks from infection. In Cameroon, the amount offered was slightly less than the poultry might earn in the market during a normal period, although was higher than the price that birds were earning once the outbreak had occurred, when prices for poultry products fell sharply.

***Culling and movement restrictions*** explains the need for zoning and slaughter of birds when outbreaks occur. This is very important information for farmers. Unless poultry keepers understand the need for these measures, they will naturally be very fearful of reporting their sick poultry to veterinary authorities, and may be hostile if culling teams come to their farms and villages in the event of an outbreak.

***Compensation – necessary for control*** gives the example of Cameroon, where compensation was given to farmers when their birds were culled. The outbreak which occurred in March 2006 was successfully controlled, not least because of the compensation offered by the government. The interview also looks at what other kinds of support are needed by the wider poultry industry, since many other businesses were negatively affected by the avian flu outbreak.

## **Using this Rural Radio Resource Pack**

Broadcasting accurate information on avian flu is extremely important, but not always easy. For example, in some countries there may only be one or two people who have sufficient knowledge of the subject, and they may be too busy to take part in your programme. However, reporting incorrect information can be worse than reporting no information, so broadcasters must be sure that their sources of information are accurate.

All the information in this radio pack has been checked by experts, and the interviewees have also been carefully chosen for their specific expertise on the subject.

The interviews in this pack focus on the essential, practical knowledge that rural people and poultry keepers need in connection with avian flu. Unlike some other subjects covered in RRRPs, this knowledge is not area-specific. In other words, all or nearly all the information in these interviews will be relevant to your listeners, no matter which country you are in.

You can build on the information in these interviews, to include country or area-specific details that your listeners will value. Here follow some suggestions for additions you could make.

### ***Recognising and responding to disease symptoms***

This interview raises the issue of whether farmers should be looking for signs of disease in wild birds. In fact, wild birds often do not show signs of avian flu, even if they are infected. An expert in your country – for example from the Wildlife Service – could comment on which areas in the country are most likely to be visited by migratory birds. This will normally be wetland areas, where birds such as ducks and geese spend part of the year. Farmers living close to these wetland areas should be especially watchful for signs of disease in their poultry, and should take whatever steps they can to prevent their poultry coming into contact with wild birds.

### ***Protecting village and backyard poultry and The importance of enclosures***

In the first of these interviews, farmers are encouraged to avoid giving contaminated feed to their poultry. This is a good subject to address in more detail. A poultry expert could comment on safe sources of feed available in your country. These might include commercially sold feeds, and locally produced alternatives.

In villages, many households keep a few poultry which scavenge for food without any restraint. In the context of avian flu, these birds are very vulnerable to the disease should it come to an area. But changing from a scavenger system to a system of keeping birds in enclosures is a big step to take. It would therefore be useful to get more information on how this can be done by talking to a farmer who has done so. For example:

- How did they go about it?
- How do they feed their poultry?
- Have their poultry become more productive, and what other advantages have they gained from enclosing their birds?

An extension officer could also comment on this subject, but getting the actual experience of a farmer will be most interesting for the listeners. It is worth noting that apart from keeping a flock free from disease, there can be many other advantages of caging or enclosing birds, for example:

- less risk that birds will be killed or taken by thieves or predators (if enclosure is well made, and secure);
- protects birds from being run over by vehicles;

- a less stressful environment, allowing poultry to grow quicker;
- more control over what poultry eat;
- less risk of contamination in food or water sources.

### ***Protecting commercial poultry farms***

Commercial poultry farms should already have strict rules to keep diseases out of the farm – known as biosecurity rules. However, farms can still be at risk from avian flu. The most likely routes of infection are:

- buying day old chicks which are already infected with the virus;
- using feed which is contaminated with the virus;
- allowing visitors onto the farm who have contaminated material (such as poultry faeces) on their shoes, clothes, equipment, or vehicle tyres.

Commercial poultry farmers will therefore be interested in learning about what procedures are being taken in your country to test chicks that are imported into the country, or are produced locally, and sold by hatcheries. How can they be sure their day old chicks are disease-free? Similarly, what advice do local experts have on clean sources of feed? And are there any other major sources of risk that are particularly important in your country? Further detail on how to quarantine sick birds, and how to introduce new birds into a flock, would also be valuable.

### ***Transmission of avian flu to humans***

To give this subject a local focus, you could invite a health professional to:

- compare the symptoms of avian flu with other diseases that people commonly suffer from in your country;
- give advice on what people should do if they think they have caught avian flu.

One important area concerns children who keep poultry, or have contact with them. In Asia, the majority of people who have died from avian flu have been children and young people. What information is being specifically targeted at children in your country, for example in schools, so that they understand the risks of contact with poultry, and how to protect themselves against poultry diseases?

In terms of cooking and eating poultry products, are there any locally eaten foods that could put people at greater risk of catching avian flu? Are there any local foods which use uncooked, or partially cooked poultry meat or eggs? Should people continue eating these? If cooked chicken or eggs are sold in street markets, how can people be sure that they are properly cooked? A health professional could offer advice.

### ***Culling and movement restrictions***

Procedures for culling and restrictions on movement of birds, vehicles and people in the event of an outbreak can vary between countries. It would therefore be very valuable to get an official to comment on what procedures will be followed in your country. For example:

- Who should farmers report any suspicious bird deaths to?
- What will the response be if an incident is reported?
- What administrative system is in place to respond to suspected outbreaks? (e.g. is the country divided into regions, each with its own response team?)
- What capacity do those teams have – are they likely to be overwhelmed?
- If an outbreak occurs, does the government plan to cull birds in the immediate area?
- Who will do the culling, and how will birds be slaughtered?
- What will be done with bird carcasses?

- What policy does the government have on zoning, and restrictions within these zones?

***Compensation – necessary for control***

Similarly, listeners will be interested to know what plans the government has to compensate farmers, should slaughter of poultry be necessary. You will probably need to speak to a ministry spokesperson to find out.

- Is the government planning to pay compensation for slaughtered birds?
- How much per bird will be paid?
- Is this an appropriate amount – is it enough to encourage farmers to report cases of sickness or death in their flock?
- Which birds will be compensated for?
- What will be the procedure for registering birds?

***Compensation - necessary for control*** describes a case in Cameroon where compensation was paid and the disease outbreak was quickly controlled. If the government is not planning to pay compensation, do they have any other ways to encourage farmers to report sick or dead birds?

This interview also looks at whether the government in Cameroon will be offering any wider support to those working in the poultry industry. There may be many businesses that were affected by the outbreak who have had no financial support. This issue could also be explored in your local context.

### **Other aspects of avian flu**

This resource pack concentrates on practical information that people need in order to detect and respond to an outbreak of avian flu, if one should occur, and how they can protect themselves against infection. There are many other issues concerning avian flu which are not covered, but could be of interest and value to your listeners. These include:

- Vaccination of poultry: vaccination is another control strategy which can be used to limit outbreaks of the disease. What is current scientific thinking on the effectiveness of vaccines? Could vaccination be an appropriate control strategy for Africa?
- International and regional efforts to control the spread of avian flu. This includes research by international and regional organisations into the nature of the virus, and the search for an effective vaccine. Also, agreements between countries over movement of poultry across borders, and regional control strategies should an outbreak occur.
- Role of trade and cross-border movement of poultry and poultry products in spreading the disease. For example, some outbreaks in Africa may have originated in chicks brought from Asian hatcheries.
- Role of migratory birds in spreading avian flu. What are the mostly common routes for migratory birds to take, which countries are most at risk, and which species of birds offer the most threat?
- Avian flu in East Asia. The H5N1 virus is now entrenched in some Asian bird populations. What has the experience of avian flu in East Asia been? What is the current situation in terms of outbreaks? What progress has been made in controlling the disease in Asia?
- Danger of avian flu becoming a human pandemic. What is the likelihood of a human to human spread of H5N1 avian flu occurring? What would be the implications if the virus mutated to a form which could spread between people?
- Informing rural communities. What are the best strategies for getting information on avian flu to people in rural areas?
- Preventative strategies. Apart from giving information, are there other kinds of support that governments or NGOs can offer to poultry keepers, to reduce the likelihood of an avian flu outbreak?

## **Further information**

### **Online articles and fact sheets**

**Man, bird and beast** is an article in CTA's Spore magazine, looking at strategies that can be used to control avian influenza and other zoonotic diseases, particularly in the developing world.

[http://spore.cta.int/spore123/spore123\\_feature.asp](http://spore.cta.int/spore123/spore123_feature.asp)

**What you need to know about avian flu** is a fact-sheet produced by the Kenyan National Task Force on Avian Flu. It can be found at the following webpage:

[http://www.ilri.org/ilripublication/uploaded%20files/Avian%20flu%20press\\_Revised%208th%20March%202006.pdf](http://www.ilri.org/ilripublication/uploaded%20files/Avian%20flu%20press_Revised%208th%20March%202006.pdf)

A **Swahili version** is available from the following link:

[http://www.ilri.org/ilripublication/uploaded%20files/Avian%20flu%20press\\_Revised%208th%20March%202006\\_Swahili.pdf](http://www.ilri.org/ilripublication/uploaded%20files/Avian%20flu%20press_Revised%208th%20March%202006_Swahili.pdf)

**Bird flu a long term threat to Africa** is an article on the BBC website reporting on the economic impacts of the disease in Africa.

<http://news.bbc.co.uk/1/hi/world/africa/4769148.stm>

**Small-scale poultry producers: falling foul of avian flu?** is an article in New Agriculturist online magazine that looks at the likely impact of avian flu on small-scale poultry keepers.

<http://www.new-agri.co.uk/06-3/focuson/focuson1.html>

### **Books**

**Small scale poultry production in the tropics** (ISBN 90 72746 58 9) by van Eekeren et al., Publisher - CTA, 1995, Agrodok 4 (5 CTA credit points)

**Poultry** (The Tropical Agriculturalist series) (ISBN 0 333 79249 5) by Smith, AJ, Publisher - CTA/Macmillan, 2001 (10 CTA credit points)

**Bird flu/Avian influenza 2006: The H5N1 virus, poultry, other animals, and people here and elsewhere, from Asia to Africa** (ISBN 2-87614-639-8), Publisher – CIRAD, 2006

**The monster at our door: the global threat of avian flu** (ISBN 1 59558 011 5) by Mike Davis, Publisher - New Press, 2005

**Everything You Need to Know About Bird Flu and What You Can Do to Prepare For It** (ISBN 1405095733), by Jo Revill, Publisher – Rodale International Limited, 2005 (Written for the British general public)

**The Bird Flu Handbook: What Is Avian Influenza, And What Do We Need To Know To Be Prepared For A Pandemic?** (ISBN141165899X) by Manoj Kunda, Publisher -Lulu.com, 2005 (Written for the American general public)

**Virus Alert! the Truth About Bird Flu!** (ISBN 1846851033) by John Piper, Publisher – Exposure Publishing, 2006

## **Useful Websites**

Up to date and accessible information on the latest developments in avian flu, both disease outbreaks and research activities, can be found in the bird flu dossier on the **SciDev.Net website**: <http://www.scidev.net>

Extensive information on avian flu outbreaks in Africa, and numerous links to other sources of information can be found on the **International Livestock Research Institute's website**, at the following webpage:  
<http://www.ilri.org/ilripublication/ShowDetail.asp?CategoryID=NEWS&ProductReferenceNo=NEWS%5F050415%5F002>

A detailed, scientific description of avian flu viruses, how they are spread, control measures and risk to humans can be found at <http://www.influenzareport.com/ir/ai.htm>

A wide range of mostly technical information on the disease from the **FAO's Animal Production and Health division** can be found at:  
[http://www.fao.org/ag/AGInfo/subjects/en/health/diseases-cards/special\\_avian.html](http://www.fao.org/ag/AGInfo/subjects/en/health/diseases-cards/special_avian.html)

Specific information on safe handling of poultry and eggs can be found at the following website:  
[http://www.who.int/foodsafety/fs\\_management/No\\_07\\_AI\\_Nov05\\_en.pdf](http://www.who.int/foodsafety/fs_management/No_07_AI_Nov05_en.pdf)

## **Contact details**

**International Network for Family Poultry Development** – the network has representatives in most African countries, who could be contacted for further information or interviews. Details of the network can be found on the following website: <http://www.fao.org/ag/aginfo/subjects/en/infpd/home.html> or can be contacted through the central office in Rome, address below.

Emmanuelle Guerne Bleich  
FAO HQ, Room C-572  
Viale delle Terme di Caracalla  
Rome 00100, Italy.  
Tel: +39 06 570 56660  
Email: [emmanuelle.guernebleich@fao.org](mailto:emmanuelle.guernebleich@fao.org)

**The International Livestock Research Institute** in Nairobi can be contacted at the following address:

ILRI  
PO Box 30709  
Nairobi 00100  
Kenya  
Fax: + 254 20 422 3001  
Email: [ILRI-Kenya@cgiar.org](mailto:ILRI-Kenya@cgiar.org)  
Website: [www.ilri.cgiar.org](http://www.ilri.cgiar.org)

**Zimbabwe National Society for the Prevention of Cruelty to Animals**

PO Box CH55

Chisipite

Harare

Zimbabwe

Tel. +263-(0)68-23519

Fax +263-(0)68-23443

Email: [conroc@wmeb.co.zw](mailto:conroc@wmeb.co.zw)

**World's Poultry Science Association**

Dr Ir Piet C.M. Simons

Secretary General WPSA

PO Box 31

7360 AA Beekbergen

The Netherlands

Phone: +31 55 506 3250

Fax: +31 55 506 4858

[piet.simons@wur.nl](mailto:piet.simons@wur.nl) or [wpsa@xs4all.nl](mailto:wpsa@xs4all.nl)

**Network for Smallholder Poultry Development**

Dyrlægevej 2

1870 Frederiksberg C

Copenhagen,

Denmark

Phone: 3528 3761

Fax: 3528 3762

Email: [poultry@kvl.dk](mailto:poultry@kvl.dk)



# CTA

## *Rural Radio Resource Pack – 2006/3*

### **Avian Flu**

| <b>Contents</b>                                                                                                                                                                    | <b>Duration</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b><i>Recognising and responding to disease symptoms</i></b><br>Signs of avian flu in poultry and what farmers should do if they suspect their birds may be infected.              | 4'54"           |
| <b><i>Protecting village and backyard poultry</i></b><br>How avian flu can spread to backyard poultry, and how farmers can prevent this.                                           | 6'43"           |
| <b><i>The importance of enclosures</i></b><br>How to keep enclosed birds free of disease, and practical advice on construction of enclosures for poultry.                          | 6'51"           |
| <b><i>Protecting commercial poultry farms</i></b><br>A commercial poultry farmer explains what he is doing to prevent avian flu entering his farm.                                 | 4'44"           |
| <b><i>Preventing transmission to humans</i></b><br>Who is at risk of catching avian flu from poultry, and what should such people do to protect themselves?                        | 5'01"           |
| <b><i>Poultry meat and eggs – safe to eat?</i></b><br>When are poultry and eggs safe to eat, and how should people handle and cook them to eliminate risk of disease transmission? | 3'12"           |
| <b><i>Reducing risks for people and poultry</i></b><br>Practical advice on observing flocks, minimising risk from bird droppings, and buying new birds.                            | 5'07"           |
| <b><i>Increasing biosecurity when outbreaks occur</i></b><br>How farmers and traders can minimise the risk of a widespread outbreak.                                               | 5'56"           |
| <b><i>Culling and movement restrictions</i></b><br>Why culling is necessary, how it should be managed, and what restrictions on movement are needed during an outbreak.            | 4'09"           |
| <b><i>Compensation – necessary for control</i></b><br>Why compensation for slaughtered birds is important, how a compensation system was managed in Cameroon.                      | 5'20"           |



## Avian flu

### *Recognising and responding to disease symptoms*

**Cue:**

Since February 2006, when the first outbreaks of H5N1 avian flu to occur in Africa were reported from Egypt, farmers across the continent have been told to be on the alert for signs of the disease. But what signs or symptoms of disease should they be looking for? Is avian flu very easy to recognise? Does it have very different symptoms from other diseases? And what should farmers do if they think that one or more of their birds may have, or may have died from, the disease?

Dr Christine Chisembele is the National Coordinator for the Avian Influenza Prevention and Control programme in Zambia. She spoke recently to Chris Kakunta, who began by asking what symptoms of disease poultry farmers should look out for.

**IN:** “Yes the first symptom that they ...”

**OUT:** “...You keep the species separated.”

**DUR’N** 4’54”

**BACK ANNOUNCEMENT:** Dr Christine Chisembele with some important advice in how to prevent poultry flocks from catching avian flu. She was speaking to Chris Kakunta, and the interview comes from a radio resource pack produced by CTA.

**Transcript**

**Chisembele**

Yes the first symptom that they will notice is that the birds are no longer thrifty, and usually it comes with high mortality. Sometimes before you see any signs birds just drop dead. Of course it is difficult for the farmer to distinguish between especially Newcastle and this particular virus, so the best right now they can do is to inform the veterinary officers, even the health centres, tell them that they are having such an experience. The other symptoms they will notice is the birds, usually their faces do swell, in certain cases, when the disease lasts long enough for them to see the symptoms. Their heads swell and they become, to use a simpler word, reddened, there is that discolouration of red, purplish look, especially on the combs and also on the feet. Egg production for layers drastically reduces. So the main symptom you see is this reddening of surfaces which have colour, and then of course if the bird is opened a lot of organs are haemorrhaging. So usually when they see such they shouldn’t start opening up birds and doing whatever else, they should quickly inform the professionals, the nearest officers that they can come across.

**Kakunta**

*You earlier mentioned that a farmer can mistake it for Newcastle. How can they distinguish the two?*

**Chisembele**

Its is very difficult for a farmer to distinguish. It can only be done through laboratory tests. So on suspicion we actually put in place measures to control any further spread of such a disease.

**Kakunta**

*Your emphasis has been on hybrid chickens and those that are meant for eggs. What about small scale farmers who keep their chickens on free range? How can they know that their birds are also affected by avian flu?*

- Chisembele** The same will apply to our free range farmers because their birds will stop running around the way they usually do. And in fact these birds unfortunately are not as closely monitored as the ones which are housed. So the farmer may just notice that birds are dropping dead around, and the farmer only has to inform the nearest veterinary or health centre, to tell them that this is what is happening to his flock.
- Kakunta** *I wonder if these symptoms also apply to wild birds?*
- Chisembele** In most cases in wild birds you won't even see the symptoms. There have been reported that in most cases they just shed the virus. In a few instances they die, they just drop dead. But of course on post-mortem these symptoms can be seen.
- Kakunta** *Now once the farmer notices the symptoms, the signs that you talked about earlier on, what should they do?*
- Chisembele** Immediately they notice such symptoms they should inform the nearest veterinary or health centre. And they should keep the poultry house closed. Keep everybody away from that house until the professionals come. If anybody has to enter the poultry house, they have to go through the disinfection process. They dip their feet in disinfectant, their hands are washed in disinfectant, and they go with protective clothes, which when they leave that house they change. Then the following actions which involve collection of samples, depopulation, and all that will follow. And that usually should be taken care of by the professionals to reduce any risk of transmission to the humans.
- Kakunta** *How quickly can this disease spread from one area to another?*
- Chisembele** Oh, it can spread very fast. Contamination comes from faecal excretions, through nasal secretions and through people and utensils. So if you are using the same utensils in all poultry houses it will spread. And you know how you find that sometimes the neighbouring farmers come to that farm without knowing that it is infected; they can carry the infection. So it is a virus that can spread quite fast.
- Kakunta** *For the sake of our listeners at small scale level what exactly should they do to ensure that these measures are adhered to?*
- Chisembele** Yes, even the small scale farmers, we are trying to encourage them to house their birds. To house them because you find that a number of them, they could be close to the water bodies, where there could be these birds that could transmit. As I mentioned, the wild birds would maybe not even show symptoms, they will just be shedding the virus, like the ducks and geese. So as much as possible we advise that the villagers house their chickens, and then they keep them according to species. Like you don't mix a chicken with a goose or a duck. You keep the species separated. *End of track.*

# Avian flu

## *Protecting village and backyard poultry*

### **Cue:**

Avian flu is known to be a highly contagious disease – it can be passed very easily between poultry species, and occasionally can cause illness and death in humans. Unlike some viruses, avian flu can also survive outside the body for a long time, for example in the faeces or other excretions of infected birds, or in water where infected birds have been drinking or swimming. When an outbreak occurs, village poultry keepers whose birds scavenge in the open can be at high risk of contracting the disease, either from other poultry, from wild birds or from infected materials such as food and water.

So what can be done to increase the safety of village chickens? Nicholas Kauta is the Ugandan Commissioner responsible for livestock health in the agricultural ministry. He spoke to Eric Kadenge about some management practices which are now recommended for village poultry keepers in order to prevent the disease taking hold in the country. He began, however, by explaining more about how avian flu can be spread.

**IN:** “In many cases people carry birds ...”

**OUT:** “...problem from coming to the country.”

**DUR’N** 6’43”

**BACK ANNOUNCEMENT:** Dr Nicholas Kauta, Uganda’s commissioner responsible for livestock health, was talking to Eric Kadenge. The interview comes from a radio resource pack produced by CTA.

### **Transcript**

#### **Kauta**

In many cases people carry birds large distances for cultural purposes, then they introduce the disease into new areas. In the case of wild birds, they fly long distances to new areas during migrations and where they stop they may get in contact with local birds, infect them, and the disease spreads from there. But there is another mode of transfer. For example, all the excretions from affected birds carry a lot of viruses, so if for example a vehicle which was carrying a bird which is infected gets the dropping of the chicken, or the fluids which come through from the nose, the beak of the bird, when it goes to new areas the droppings can still start a new infection in a place, when the birds of that new area contact it. And the other thing is to move the poultry products, like the meat; even eggs get infected. And you can move eggs to new areas and cause problems. You can also transfer through trays used for carrying eggs. So all those are avenues of moving the disease from one place to another.

#### **Kadenge**

*Given then that there is a high chance of the virus spreading as you have just mentioned, what should especially backyard poultry farmers do to ensure that they avoid or reduce the risk of their poultry being infected by the avian flu?*

- Kauta** You address the sources of feeds. You may keep your birds away from other birds but if you bring contaminated feeds you have not done well. If you bring contaminated water, same problem. So the food you bring should be from a good source which is hygienically recommended. The water you provide to the birds should not simply be drawn from the running waters around where ducks are playing, everything is playing. Maybe because of this disease we should even learn to boil water for birds, just to save them. Now the other thing is preventing contact between other birds and your birds; that calls for starting to change management practices, because the backyard and free range system is the most risky if the disease comes. So we probably have to shift away from the traditional methods, and start enclosing our birds. To me that is the only way out: changing in management practices.
- Kadenge** *Now when you mention the practices, I would like to get to know what are some of the things that you must make sure, like you mentioned earlier the eggs.*
- Kauta** In this country we have a problem in that the birds live as we call it traditionally, meaning the most primitive method of keeping them. In other countries, birds are kept in age groups. One age group lives together until it is sold. And they come from known hatcheries; they don't mix birds from different sources. Here, in a single home you may have birds coming from so many different sources. Those brought as gifts, those bought from the market, those that hatch there, and all those are recipes for very, very serious problems. So, when we talk of management practices, we are saying, be sure of where you get your birds from. Or even mixing birds, there is a danger that some birds are carriers. Like the ducks tend to be carriers of this disease. They may have it and not show it as quickly as the chickens. So you may introduce a duck and your chickens die earlier than that duck. So we are saying don't mix species, keep one species at a time. If it is chickens, chickens. Don't mix them with turkeys, don't mix them with guinea fowls, like we tend to do. If the eggs don't hatch, what is the reason they don't hatch? We have a habit of just throwing them out. They are better buried, because they will have a high concentration of viruses, and that can be a source of more problems.
- Kadenge** *Do you think it will be easy to handle cultural issues, like someone getting a chicken from a relative and bringing it into the compound?*
- Kauta** That is not an issue which is solved by ordering people around, or planting regulations. The people have to appreciate by themselves that what they are doing is not good, it undermines their poultry sector. Because whoever carries a chicken to a friend simply carries a very small thing and it goes, you cannot enforce it. So something you cannot enforce, you only bank on giving people the relevant information to take meaningful decisions, and that's it. If the people don't appreciate it then the disease will be impossible to control should it break out.
- Kadenge** *Now after you have observed all the hygiene that you could on your farm, there comes a time when you need to go to the market, and that could be another area where your birds could be exposed to the virus. Let's say you did not sell all of them. What do you do when you come back home with some left?*

- Kauta** Under the circumstances of avian influenza you cannot take birds back to your farm. There is a need to organise the marketing system so that birds that are not sold remain in the market areas. They may be sold the next day or the other day, but never returned to their home. Because the market is the meeting place for all types of diseases. And even if you don't get avian influenza you will get something else, like Newcastle disease, infectious bronchitis, there are many diseases affecting poultry. So it is not recommended at all to return a single bird back home after it has reached a market.
- Kadenge** *And to wind up this interview, is there anything else you would like to add to a farmer in regards to the issue of avian flu?*
- Kauta** The farmers know very well that almost every year there is a wave of death of birds associated with different diseases. Now these diseases are different. Although they kill birds in mass, this disease has a potential to wipe out the entire population and affect human beings as well. So it is real threat and it must be taken seriously. Let us change management systems. Even if you have only ten birds get a shelter for them, feed them from there, that we prevent the problem from coming to the country. *End of track.*



# Avian flu

## *The importance of enclosures*

### **Cue:**

The H5N1 avian flu virus is not the first kind of avian flu to affect poultry flocks, either in Africa or other parts of the world. However, this particular strain of the virus is especially dangerous, because as well as affecting birds, it can also be passed to humans, and is often fatal. Once an outbreak occurs, the most effective way of preventing it from spreading is, currently, by destroying affected birds and preventing transfer of the virus outside the outbreak area. However this is an expensive and labour-intensive process. It is therefore much better to invest in protecting poultry flocks so that they are less like to catch the virus, than try to deal with outbreaks when they occur.

So how can poultry, flocks be protected? For some diseases such as Newcastle disease, vaccinations are available. However, with avian flu the effectiveness of vaccines is less clear, and most countries are not adopting vaccination as a control strategy. Instead, poultry keepers are being advised to protect their flocks by housing their birds in some kind of enclosure or cage, and taking precautions to ensure that all food and water are disease-free.

Dorothy Tumbo lives in Bulawayo in Zimbabwe, and keeps chickens in an enclosure, called a fowl run, in her backyard. She spoke to Sylvia Jiyane about how this helps to protect her birds, and began by explaining how, on one occasion when some of her birds got sick, she took them to a local veterinary clinic. The vets feared they might have a form of avian flu, although not the H5N1 strain, which has not yet occurred in Zimbabwe.

**IN:** “At one time I had about 50...”  
**OUT:** “...more if they are infected.”  
**DUR’N** 6’51”

**BACK ANNOUNCEMENT:** Celia Abolnik, a senior research assistant from a veterinary institute in South Africa, was talking to Lucas Moloi. Earlier we heard Sylvia Jiyane interviewing urban poultry keeper Dorothy Tumbo. The interviews come from a radio resource pack on avian flu produced by CTA.

### **Transcript** **Tumbo**

At one time I had about 50 that I had taken, and over 30 actually died because of avian flu. What actually happened is that the other person, who was about five houses away from where I was, had their chickens actually dying. And then mine also started dying and when we took our chicks to the vet, that was when we were told there was an outbreak of a disease actually around our area. They thought it was avian flu, and they also suspected it could be Newcastle, but we left them with the chicks so they could actually test. These guys, the people from the vet, also encouraged us to clean our fowl runs. If one group of poultry is ready for market you remove it, then you have to clean your fowl run and spray it with chemicals before you put a new batch of poultry into the fowl run. You find that if you keep on doing that you find that cleaning, you have to be clean, you have to spray your fowl runs, and you have to make sure that the water and all the vaccines are given to the poultry, and also make sure that the eating utensils are clean, and there is a clean supply of water. That way you eliminate the diseases, but if you are not very careful then you will see that the chicks or the poultry will actually start to just die and if you don’t realise it at the end of the day you won’t have anything left, because all of

them will be attacked by the diseases.

**Jiyane** *That brings us to the point of access to the poultry house. In your own experience, maybe at your house, who normally accesses the poultry enclosure?*

**Tumbo** Actually I have got one person who is managing that, sort of a worker, a woman I employed, and she's the one who is actually feeding those chickens. You look at the number of chickens you are having, and then you have to make sure that you have enough feeding trays that will cover, so that at least you don't keep on going into the fowl run now and again to replace the feed that will be finished. What you have to do, you have to look at them well measured, to say that at least if I've got these big, five or six big containers, and if I put in the morning, it means the next time I will put the feed will be in the evening, so that at least you limit the number of going in and out to about twice a day only, and with only one person going in to replace the water the feed, and everything. So you have to have feeding trays and water points which are almost enough to last a full day, or about 12 hours, before you go in there.

**Jiyane** *And besides the risks of diseases, like we have mentioned avian flu, Newcastle and other diseases, what could be the other risks of keeping poultry in the backyard?*

**Tumbo** Yes the other risks, especially in urban areas is there are a lot of thieves, and if people realise that there is poultry that is being kept there, they would want to come and actually steal them when they are almost ready. So your fowl run has to be quite strong, and if possible put some burglar bars, and also make sure that it is always under lock and key so that people don't access your chickens. Otherwise, besides the thieves the other risk is mainly that one we have talked about of diseases. Because if all those are eliminated it is quite a viable business which can actually generate a lot of income for people.

**Jiyane** *And finally, Dorothy, in the event of a disease outbreak in poultry enclosures, what are the practical steps that one should take to prevent the disease from spreading?*

**Tumbo** If you have noticed that, especially maybe the first thing you have to see is maybe you can see one animal dropping dead, and then you see another maybe showing signs that it is not well. What you have to do is you have to remove them, actually isolate them, remove those birds that you see are not well. Remove them and put them somewhere away and actually observe. If it keeps on going then you have to seek advice from other people, like what I said in the beginning that we ended up taking the birds to the vet, so that at least they can test to see what it was that was affecting the chickens.

- NARR** *While keeping backyard poultry in enclosures may be possible for urban poultry keepers like Dorothy, what about for those in villages, whose chickens normally scavenge for food around the homesteads? Should they also be kept in an enclosure? In areas where the risk of avian flu is high – for example because of the presence of migratory birds – keeping all poultry in some kind of house or enclosure is definitely recommended. To find out more about how, and why this should be done, Lucas Moloi spoke to Celia Abolnik, a senior research assistant at the Arc-Onderstepoort Veterinary Institute in Pretoria, South Africa.*
- Moloi** *In areas affected by the avian flu, what are the advantages of keeping poultry in fenced enclosures?*
- Abolnik** Obviously if your birds are fenced they are separated from birds that are free ranging that might be carrying infectious diseases. In the ideal situation you would want an enclosed structure, but the best thing to do is not to leave open food and water that will attract wild birds into contact with chickens. So the sharing of food and water is probably the major way in which wild birds, or free ranging birds are encouraged into contact with poultry.
- Moloi** *So is there a particular material and design to construct a safe enclosure?*
- Abolnik** It must be bird proof, it must be rat proof. The water must be, and the feed must be separated from any contact with birds and rodents from outside.
- Moloi** *Some poultry keepers may not afford expensive material and design. What should they do to protect their poultry?*
- Abolnik** You could use inexpensive materials to build an enclosure that is rat and bird proof.
- Moloi** *Like?*
- Abolnik** Mud bricks, grass, sheet metal, anything that they have access to.
- Moloi** *Should there be restrictions on who to enter poultry enclosures in areas affected by the avian flu?*
- Abolnik** Definitely. A poultry owner should not allow any person just to walk into his poultry enclosure, especially since they may be carrying infected poultry or wild bird faeces on their person, their boots, their shoes, their clothes, their hands, and handling of birds might transmit the disease in that way.
- Moloi** *Is there a special diet that poultry could be fed with to reduce chances of contracting the avian flu?*
- Abolnik** There is no special diet that will prevent them getting avian flu, although the importance of good nutrition is vital, because birds that are immuno-suppressed, birds that have a very weak immune system, sometimes due to hunger, or because they are not getting all the nutrients, might be more predisposed to infections, and might suffer and die more if they are infected. *End of track.*



## Avian flu

### *Protecting commercial poultry farms*

**Cue:**

When avian flu first broke out in Africa, the outbreaks occurred not in backyard birds, but in commercial poultry flocks. For the owners of large poultry farms, the impact of an avian flu outbreak is devastating. Not only are all their birds slaughtered, but often the public loses confidence in poultry products and the whole poultry sector is affected. Keeping commercial poultry farms free of disease, by tight controls and good hygiene practices is therefore essential.

In Malawi the government has established three avian flu rapid response teams, for the three regions of the country. It has also run training workshops for those involved in rearing poultry. Mr Jampa Banda, owner of Tiyanjane Poultry farm near Blantyre, has attended several such workshops. He told Patrick Mphaka about what he is now doing to prevent the disease affecting his farm.

**IN:** “In general, avian flu can be prevented ...”

**OUT:** “...and that will be terrible.”

**DUR’N** 4’44”

**BACK ANNOUNCEMENT:** Mr Jampa Banda, a commercial poultry farmer, was telling Patrick Mphaka about the measures that have been taken to protect his farm from the risk of avian flu. The interview comes from a radio resource pack produced by CTA.

#### Transcript

**Banda** In general, avian flu can be prevented through general cleanliness starting from the infrastructure itself, that’s the cages, as well as the personal hygiene involving all those individuals working in the poultry farm.

**Mphaka** *What changes can I see now if I am to visit and see some of the cages around, see how the chicks are doing?*

**Banda** There are several things which have been implemented. One is the change in the strategy in terms of raising the flocks. Initially we used to have a single flock mixed with the other flocks, may be younger ones, but now, we have to sell one lot before we introduce another one. That’s a major change which is obviously affecting our production, since now we are taking much longer to produce one lot of a flock. Again, we have improved the general hygiene around the poultry farm whereby cleanliness is now taken as a priority to make sure that we do not contaminate the flocks. And another measure which has been implemented is making sure that the individuals working in the poultry farm don’t keep any poultry at their homes. Now it is like a serious measure which has to be checked. So it’s a general consensus that each worker who is involved in taking care of the flocks should never keep any poultry at his home. And we are also restricting any passers-by to make sure that they do not contaminate our flock or the farm at large. So it’s like a restricted site, much more serious these days than ever before.

- Mphaka** *It must be a big challenge, I suppose, for the workers whom you have because I have seen that as a big poultry farmer, you have some of your workers who stay within the farm itself, they have their houses there, and most of them must have been keeping some poultry themselves. Was there a party where they had to slaughter all their chickens?*
- Banda** Not necessarily a party, but it was a general understanding that if we do not implement these measures, then we both lose; I as the farmer, as well as the workers themselves, eventually we were going to end up both losers.
- Mphaka** *I suppose it must be very hard to follow such rules when you know that avian flu is just something we hear about, it has not hit Malawi yet. How faithful or committed can people be getting such instructions for something they have not seen?*
- Banda** Knowing how dangerous this avian flu can be, then no chances can be spared. As such, each one of us understands that once we are hit, then there will be no escape, that is why we have taken this disease very seriously and we do not want to take any chances.
- Mphaka** *When we were coming in here, before entering, you made sure that I removed my shoes outside and then you disinfected me. How can that help, knowing that there are birds all over around and if any one of them is contaminated or infected, they can come right inside your farm?*
- Banda** There are some measures by which we make sure that the disease never gets into the farm. Some of them are obviously to make sure that our flocks are always kept enclosed in their cages. And we also keep on checking if at all there are any intruding birds coming in; and those are obviously chased out before they can even have a chance to contaminate our flock. But if at all we see any suspicious bird, maybe being weak, or whatever, we make sure that it is immediately quarantined and a veterinary officer called to make sure that he assists and check whatever is happening.
- Mphaka** *Do you believe it will ever come to Malawi, looking at the measures which some of you have taken up and have instituted?*
- Banda** It's a tricky situation. We definitely do not want to take things for granted and we can not cheat ourselves by saying it will never come here. It was introduced in other areas, therefore, it can as well be introduced in Malawi. That is why we are taking every opportunity to prevent it from coming whilst making sure that if possible, even if it comes, but at least my farm should not be affected by making sure that we implement that advice which has been given by the government. If it ever comes, then it can definitely be a disaster because apart from losing the market, obviously the economy will be affected since people will definitely be terrified and they would not buy even those chickens which are not affected, and that will definitely be very bad for everyone, loss of jobs, loss of business, and that will be terrible. *End of track.*

## Avian flu

### *Preventing transmission to humans*

**Cue:**

There are many different strains of avian flu, which cause death and sickness in poultry around the world every year. However, the H5N1 strain, which is causing such concern to poultry producers and veterinary authorities, is different from others, because it is the first strain we know of that can be passed from birds to humans. So far over one hundred people have died from H5N1, mostly in South East Asia. However, scientists fear that if more human cases occur, the virus could develop into a new form which could be spread from one person to another. This could lead to a serious epidemic, causing thousands or even millions of human deaths.

Preventing transmission of the virus from birds to humans is therefore very important. And there are a number of simple measures that farmers and poultry keepers can take to reduce their risk of contracting the disease. Dr Linus Chimangha, an inspector from the Cameroon ministry responsible for agriculture, spoke to Martha Chindong about how spread of the disease from poultry to humans can be prevented and what symptoms people should look for if they think they have been exposed to birds that may have avian flu. Martha began by asking which types of people are most at risk of contracting the disease.

**IN:** "First the farmers, the poultry farmers ..."  
**OUT:** "...for confirmation of the diagnosis."  
**DUR'N** 5'01"

**BACK ANNOUNCEMENT:** Dr Linus Chimangha with some advice to people if they think they could be at risk of having contracted avian flu. The interview comes from a radio resource pack on avian flu produced by CTA.

**Transcript**  
**Chimangha**

First the farmers, the poultry farmers and their families, who have direct contact with the birds on a daily basis. Then you have veterinarians and other technicians who visit these poultry establishments for no matter what type of intervention. Then you have even journalists, journalists like you sitting here. When an outbreak occurs you will visit it and want to have in depth information so you can better inform the population. In cases where there are outbreaks you have people who are in charge of clearing the poultry houses, people who will do slaughtering of the poultry and disposing of them. Groups of people who will go in to disinfect the poultry houses. You have even researchers and laboratory technicians who will go in to collect blood samples for examination. So these are the classes of people who are very highly susceptible and more vulnerable to the infection by, not only avian flu, but equally all other epidemics which may occur in livestock or poultry.

**Chindong** *In the case of avian flu, what of small scale farmers who stay in their houses with fowls?*

**Chimangha** These are very susceptible groups of people, and it is not advisable for people who keep chickens to be living with them in the same type of establishment.

- Chindong** *So for these people at greater risk, how can they protect themselves?*
- Chimangha** Implementing some biosecurity measures, hygienic practices, whenever they touch poultry they should wash their hands well and their clothing, they should wear protective clothing, gloves and a mask, particularly in places where an outbreak has been reported. And they should have things like dips, footbaths at the entrance of the poultry house, so that they dip in their feet when they are going in and when they are going out. And then ensuring that all poultry products which they consume – in fact this one is even a general measure – should be well cooked. As you know the virus does not withstand cooking at boiling of 60 degrees Celsius for five minutes, and at 100 degrees for just one minute the virus is destroyed. So these are some of the things they should do. And equally, in case of any suspected cases, say a sick bird, they should report it immediately to the veterinary authorities for measures to be taken.
- Chindong** *When a farmer comes in contact with the dead bird, is he at risk too?*
- Chimangha** Sure, certainly he is at risk, and that is why he is not supposed to pick up that dead bird with his empty hands. He should wear gloves, or if no gloves are available he should use something like a plastic bag, and then pick up the dead bird. And you either dispose of it, or where the disease is suspected you report it to the veterinary authorities. Such birds should really be burned or buried very deeply.
- Chindong** *When human beings contract the disease, what are the symptoms?*
- Chimangha** When human beings get this disease, you know it is a disease of the respiratory system. But the pronounced symptoms start coming after some few days of incubation, say about two days to one week, at times it can even stay up to two weeks. And then you see the symptoms coming up, which just present as the common symptoms of flu or cold, with a temperature of up to 38 at times even 39 degrees Celsius. You have a painful or sore throat. Then muscle and joint pains, respiratory signs like cough, then running nose, coupled with headache, loss of appetite and occasional nausea and vomiting. Then such respiratory signs may become more severe and they could give rise to bronchitis, sinusitis, and other bacterial pneumonias. Then specifically in elderly people, or people with other infections which compromise their immune system, like HIV/AIDS, such symptoms may become even more severe and lead to death. But generally death from this disease is rare, except if it is aggravated by other underlying medical conditions.
- Chindong** *So if we notice such a case, what is the immediate thing a farmer should do at home, even at home or anywhere, what is the immediate thing that can be done?*
- Chimangha** When avian flu is suspected in any severe cold conditions which are not immediately responding to common treatment, you should rush immediately to the hospital and meet the medical authorities. These are the ones who are supposed to do more in depth diagnosis and then see what treatment they can undertake. But again the case should be reported to the veterinary authorities, so that they come in with their own investigation for confirmation of the diagnosis. *End of track.*

## Avian flu

### *Poultry meat and eggs - safe to eat?*

**Cue:**

When outbreaks of H5N1 avian flu have occurred, whether in Asia, Europe or Africa, one result is that people become very frightened of eating poultry meat or eggs. In many cases this fear is not justified. For example, people who are living in areas where no cases of avian flu have been reported should not think they will be at high risk of catching the disease. While there may be a small risk, carrying out normal hygienic practices like washing hands, knives and work surfaces after cutting up a chicken, and ensuring that the meat or eggs are properly cooked, will eliminate any risk of contracting the disease, even if it should be present.

However, in areas where an outbreak has occurred, consumption of poultry products is not a good idea, and will be prohibited by quarantine regulations. This is because even though cooking will kill the virus, people are at risk of contracting it from doing other tasks like plucking and handling the uncooked birds. Dr Philemon Wambura, of the Animal Diseases Research Institute in Tanzania spoke to Lazarus Laiser about the risks from consumption of poultry and eggs. Lazarus began by asking whether it was safe for people to eat birds which look healthy, if they come from a flock in which other birds have shown symptoms which could be avian flu.

**IN:** "No, no it is not safe because ..."  
**OUT:** "...precautions measures of whole hygiene."  
**DUR'N** 3'12"

**BACK ANNOUNCEMENT:** Dr Philemon Wambura, advising that farmers and poultry keepers maintain good hygiene standards when handling chickens, to minimise the risk of catching avian flu.

**Transcript**

**Wambura** No, no it is not safe because by just merely looking at the chickens you cannot really tell sometimes chickens which are incubating the disease. So once the flock has been infected normally there is a quarantine. So both chickens, eggs and all egg products are not really safe for human consumption or even to catch them and to be closer to the farm.

**Laiser** *How about eggs from birds from that farm?*

**Wambura** Even the egg products or everything that is concerned with chickens even the droppings and all the surroundings are not really safe.

**Laiser** *Let us think about the preparation of preparing and cooking the poultry. Are there any extra precautions that people should take when handling and preparing poultry for cooking?*

**Wambura** Yes if they are coming from clean flock you take precautions as usual, that you should be prepare your chicken under hygienic conditions. Make sure that you wash the chicken very well and you cook it very well as usual, and wash your hands after the cutting the chickens and all the hygiene measures. Just simply washing hands with soap can suffice.

**Laiser** *In terms of cooking is there any advice people should know in order to reduce the risk of them catching the virus?*

- Wambura** The big risk is from the flock itself. If the flock is clean and it has been confirmed by the authorities that your flock is ok and you proceed with the selling of meat, I think the normal cooking as usual is really safe. So people should not turn away from eating chicken because of the scare of bird flu. As long as the flock has been free from the disease and there is no other confirmation of the disease around, people should take all the precautions, measures of cooking chicken in hygienic conditions. No extra care is really needed, just normal usual cooking until it is really cooked very well, it is no problem at all.
- Laiser** *What else Dr Wambura would you like to tell the listeners, who are listening to this programme right now?*
- Wambura** I think it is very important to be aware that this disease is normally a bird disease or chicken disease. So we have to be aware of deaths, any suspicions of the disease you have to report it to the authorities, veterinary inspectors and veterinary officers or people who are concerned with welfare of people in all levels from the villages up to the nation, we have to be aware. In those countries where the disease has been reported, normally children are really affected because they play around with dead chickens and they are infected. So we should be really advising children once they see any dead birds or any suspicion of sick birds, they should not really touch them and we should inform the authorities rather than taking for granted that this is a normal death. So any suspicion should be reported and the people should not worry so much about the disease. The only thing you should take precautionary measures when you handle chickens in your normal flock, make sure you wash your hands and take precautionary measures of whole hygiene. *End of track.*

# Avian flu

## *Reducing risks for people and poultry*

### **Cue:**

Each year, the coming of winter in the Northern Hemisphere causes millions of birds to fly south, searching for warmer weather. Many end up in Africa, often settling in wetland areas. For African countries this also brings potential danger. The H5N1 strain of avian flu has been detected in several European and Asian countries, and could potentially be brought to Africa by migrating birds, such as ducks and geese. For commercial poultry farmers and small scale village farmers, an outbreak of avian flu could be devastating; a serious outbreak could also have significant impact on national economies for some countries.

In 2006, Senegal was one of several African countries where an outbreak of H5N1 avian flu occurred. This is the type of flu that can also be spread to humans, and has led to over one hundred human deaths in East Asia. The small country of The Gambia is surrounded by Senegal on three sides. So far it has remained free of the virus, but the government has set up rapid response teams in 6 zones of the country, to deal with an outbreak should one occur. So what should farmers be doing, both to minimise the risks to their poultry and to themselves? In this interview, Dr Demba Jallow, head of the national avian flu response team, offers some advice. Putting the questions was Ismaila Senghore.

**IN:** “One, try to be observing your birds...”

**OUT:** “...marginalised and the poorest therefore.”

**DUR’N** 5’07”

**BACK ANNOUNCEMENT:** Ismaila Senghore was speaking to Dr Demba Jallow, Assistant Director of Livestock Services for The Gambia, and head of the national avian flu rapid response team. The interview comes from a radio resource pack produced by CTA.

### **Transcript**

#### **Jallow**

One, try to be observing your birds everyday to observe any mortalities or sickness and other things. That’s one. And report immediately to the nearest veterinary personnel in case you notice some mortalities or some sickness in your birds, that’s one. Two, dead birds should be buried deep or burned. They should not be eaten, and any poultry products should be well cooked. Never eat sick birds. And then three you should always wash your hands thoroughly with soap and water, to kill the virus. So that is some of the advice we always give to our farmers.

#### **Senghore**

*Now Dr Jallow, we understand that the most infectious part of the poultry is the droppings, poultry droppings. Now this can be in the poultry yard or it can be in transportation vehicles, or receptacles where these flocks may be put. What do you advise people to do to make sure that they are not easily in contact with droppings which might be infected?*

- Jallow** We have always told our farmers, any time you visit your poultry, wash your hands properly, and then every time you visit the poultry, always remove all your clothes and wash them, of course your shoes and all those things. And three, when it comes to droppings, droppings should be well cleaned and then packed in one area of the farm, preferably dried and then buried. Because the virus lives best in dung which is very moist. So we always discourage moisture being in the dung.
- Senghore** *But I'm sure in some places they will recommend cleaning of all utensils and so forth, but there are areas where there might not be enough water. I'm sure fortunately here in The Gambia we seem to have adequate water. But just in areas in African countries who are also just as concerned as we are, where there is no water, what do you recommend for them to do?*
- Jallow** Now that will be very difficult. In The Gambia as you just stated, we have a lot of water. But in areas where perhaps water is scarce, perhaps they have to dry the equipment or their clothing in air, in the hope that the hot sun will sterilise this equipment.
- Senghore** *Now when you come to generally the hazards, what are the main possibilities for the virus to transfer from birds to humans?*
- Jallow** From birds to humans. Of course it is very high, especially where there is no hygiene, or where there is ignorance of the mode of transmission of the disease. That is why we always advise that you should maintain hygiene at all levels. That's why when you're cooking meat, poultry meat, etc. and when you are dealing with poultry, your poultry house should be far away from where you live. It should be well cleaned daily, and you should always put on a mask when you are cleaning the dung and all those things, and you should always wash your hands thereafter, and of course always cook your meat or eggs thoroughly. And then of course, at a commercial level, we advise farmers to maintain biosecurity so that at least there is least movement between birds from their farm to an outside farm, that's one, or birds within their farm and these other migratory birds that come to seek for feed and other things. So hygiene is the key word in the effort to prevent poultry to human transmission.
- Senghore** *And I'm sure you also want to have a say on regional movements, because there could be cross border movement of animals from other countries into our country.*
- Jallow** Exactly, just as migratory birds don't have boundaries, they move from Europe to here, they can get to our neighbouring country like Senegal and then still move to our wetlands here, without warning or so. So it is always very important, that's why you have to have regional networking with our neighbours. That we have already established with Senegal, so that we can be sharing information in terms of movements and all those things.
- Senghore** *But what about live birds, I mean poultry?*

- Jallow** Live birds, this concerns the fact that our borders are very porous and we have these weekly markets, where people across the border trade in live poultry. So this is a major source of concern. But again, just as I have told you, we have sensitised our people sufficiently enough, to be able to at least hopefully be responsible for their own well-being, or the well-being of their animals in case there is an outbreak, or even before there is an outbreak. Maintain hygiene, don't buy sick birds, don't carry sick birds to your home, always isolate them or quarantine them for at least two weeks before you introduce them into the flock, don't accept birds which are gifts and other things, especially when you think they are sick, and other things. Things like that we have well informed our people, and we hope that they will abide by such things and then minimise the risk of bird to human transmission.
- Senghore** *Do you think it is important to have a clause that for example prohibits the introduction of infected poultry into particular areas, and if anyone contravenes it, for example.*
- Jallow** This is in the Animal Disease Act of this country, that in the case of an outbreak, or the case of transboundary diseases, nobody is allowed by law to trade in such animals or birds. So anybody who is found doing such a thing can land himself in prison also.
- Senghore** *Now to conclude, what is the overall threat of bird flu to a small economy like The Gambia?*
- Jallow** It may be devastating, because up to about 80 per cent of our population are the farming community and most of them keep poultry, and these poultry help them in the solving of their small problems on a daily basis. So where poultry are wiped out then their livelihoods are destroyed and then that means that they have to go back to zero again and start from scratch. So it is actually a big threat. It can destroy a whole economy, concerning the fact that the affected people will be the most marginalised and the poorest therefore. *End of track.*



## Avian flu

### *Increasing biosecurity when outbreaks occur*

**Cue:**

Cross-border trade in poultry and poultry products is big business in much of Africa. However despite being a good source of income for the traders, and a valuable source of meat and eggs for consumers, there can be risks in this cross-border trade. Spreading disease from one country to another is one risk. For example, when Nigeria first experienced outbreaks of highly pathogenic avian flu in early 2006, neighbouring countries became very concerned. For this reason, when outbreaks of contagious livestock disease occur, governments can decide to impose a ban on all imports of the livestock concerned, at least until the outbreak is controlled.

However, few governments are can make import bans completely effective; they lack the trained staff to do it. Controlling movements of livestock within a country is even more difficult. But controlling diseases like avian flu is in everyone's interest, and it is everyone's responsibility to prevent the disease spreading. Farmers, for instance, must know the right thing to do if they suspect one or more of their birds has caught or died from the virus. In a recent interview, Excello Zidana spoke to Michael Nkosi, a lecturer at Malawi's Natural Resources College, and began by asking what farmers should do if one of their birds shows symptoms of the virus.

**IN:** "I would advise that the ..."  
**OUT:** "...Thank you very much."  
**DUR'N** 5'56"

**BACK ANNOUNCEMENT:** And asking the questions was Excello Zidana. The interview comes from a radio resource pack on avian flu produced by CTA.

### Transcript

**Nkosi** I would advise that the first thing that farmers should do is actually to report to the nearest veterinary station. That is the first thing, so that maybe officers should come, verify the signs and symptoms, if there is need undertake sampling, take birds to the lab and then they would do analysis. But unfortunately in Malawi we may not have the capacity to test for avian influenza. But the other thing that farmers need to do is to intensify bio-security measures. They should not just allow their workers to move from house to house if they have several units of birds. Maybe they should allocate specific attendants to a specific house so that the virus is not passed on to safe birds.

**Zidana** *How does the virus spread?*

**Nkosi** The virus can spread in many ways. It could be from inanimate objects like farm equipment. If you just borrow feeders from any how, drinkers from any unknown farm you can spread the virus. From clothes of people, boots... So I should emphasise that protective wear is very important and should be specific to a unit.

**Zidana** *Let's look at the handling of the sick birds, how does a farmer handle sick birds from his farm?*

- Nkosi** One, he should first of all isolate the sick birds, confine them separately. But at the same time he should put on protective wear, even a mask if it is available, because you never know, he maybe working with a strain of avian influenza virus that can affect man. So you need to be protected whether it is confirmed or not.
- Zidana** *And if this farmer wants his dead birds to be tested from any lab around, how does he carry the dead bird?*
- Nkosi** He can put the birds in a plastic bag that is having no holes or double plastic bag, the whole live bird push it there and then transport it to the lab. As I said earlier on that in this country we do not have the capacity to examine and diagnose for this virus. Then the lab knows how to safely package international samples to referral labs.
- Zidana** *Now let's turn to disposing of the dead birds. How should the process of disposing of the dead birds carried out?*
- Nkosi** If it is by mass slaughter, if it is the government policy to say we want to control the outbreak by mass slaughter, one, you can bury very deep: two metres deep, three metres. Or you can dig a shallow pit and then push all the birds there, put diesel and then burn.
- Zidana** *Why three metres Mr Nkosi?*
- Nkosi** Because some carnivores like dogs, cats, jackals, they can jump into a shallow pit, lift the animal and then they can scavenge and spread the virus.
- Zidana** *People who know avian flu very well have indicated that the virus can survive in birds' droppings. What measures should be taken when using items such as cages, bicycles or even vehicle tyres trying to go to an area where people suspect that there could be this outbreak?*
- Nkosi** The first thing is to institute wheel baths just outside the farm. You need to have a wheel bath that has disinfectant and as the vehicle is moving across all the manure will remain there. But at the same time you need to spray the whole vehicle canopy, even the tyres, so that you do not take the manure. Because you never know, you can carry the virus. About the facilities like cages, you need to decontaminate and remove all the manure if it is there, so that you move a safe product.
- Zidana** *In countries like Malawi, Zambia, Uganda, Kenya, people know that birds migrate from one region to another. Are there special rules to guard maybe the movement of wild birds from one country to another, as a region or as a continent?*
- Nkosi** That is a very tricky question. You know natural birds, you cannot govern them, but you definitely know the routes that they take from one continent to the other. The only bet is actually to avoid contact of domesticated birds with the wild birds along the shores of Lake Malawi or any other ocean in this region. We need to advise our farmers, maybe to intensively rear birds that are domestic to avoid that contact.

- Zidana** *Lastly how or what do you advise people involved in business of poultry, cross-border business people trying to bring in maybe birds or meat from birds? What do you advise them to do if they are doing that in the event of rumours that there should be avian flu within the region?*
- Nkosi** The only advice that I can give is they need to follow proper protocol. Get all the licenses for importation. They should desist from smuggling because avian influenza can not only affect their consumers; even themselves if it is brought into this country, we are all at risk. So what we need to do is desist from smuggling or black marketing and then follow proper channels.
- Zidana** *That was Mr Michael Nkosi who is an Animal Scientist at Natural Resources College, thank you Mr Nkosi.*
- Nkosi** Thank you very much. *End of track.*



# Avian flu

## *Culling and movement restrictions*

### **Cue:**

If an outbreak of avian flu occurs ordinary people will have a key role to play in controlling it. For example, if farmers notice that their birds have symptoms that could be avian flu, they will need to report this to local veterinary or health centres very quickly. Failure to report could allow a serious outbreak to occur, leading to a much higher number of bird deaths. In Zimbabwe, the National Society for the Prevention of Cruelty to Animals, or ZNSPCA, has launched a publicity campaign, to explain the dangers of avian flu to rural and urban populations. This has involved having public talks in all three national languages and distribution of leaflets and posters.

One important subject is the measures that will be needed to control an outbreak. For example, all poultry in an outbreak area will need to be killed, a process known as culling, and the movement of animals, people and vehicles in affected zones will be restricted. Helping people to understand what will need to be done, and why it is necessary, is vital.

Glynis Vaughan (*pron. Vorn*) is the Chief Inspector of the ZNSPCA and a member of the national avian flu task force. She spoke recently to Busani Bafana, who began by asking her to explain the culling process, and which birds would need to be culled if an outbreak of avian flu occurred.

**IN:** “Culling is the humane method ...”  
**OUT:** “...as well as vehicles and poultry.”  
**DUR’N** 4’09”

**BACK ANNOUNCEMENT:** Glynis Vaughan (*pron. Vorn*), Chief Inspector of the ZNSPCA, talking to Busani Bafana. The interview comes from a radio resource pack on avian flu produced by CTA.

### **Transcript**

**Vaughan** Culling is the humane method of destroying infected poultry. By ‘infected’ we mean both sick and healthy birds. When a farm is infected there are zoning areas, and all birds that fall into those affected zones are humanely destroyed.

**Bafana** *Why is it necessary to kill birds that appear healthy?*

**Vaughan** If those birds that appear healthy fall into an affected zone that means that they are still carrying the virus, which means that they are highly contagious and can spread it further. So all birds in that zone have to be destroyed to contain the virus in an infected zone.

**Bafana** *I would want to find out, who is responsible for culling: the authorities or the farmers?*

**Vaughan** ZNSPCA are tasked for the whole culling process and will be assisted by qualified personnel in the form of veterinary surgeons, as well as ZNSPCA staff. All ZNSPCA staff will be trained and qualified in the culling. According to the Zimbabwe task force preparedness plan, all culling will be done by the authorities, not the farmers themselves.

- Bafana** *What are the best methods of culling which are also humane and effective?*
- Vaughan** The best proven methods for culling that have been approved round the world, are the use of a mixture of carbon monoxide and argon, and gassing the birds, which is the most humane and effective. Obviously unfortunately this is only practical in large scale farming practices, but that is the best method of culling. And then also the mechanical method which you will use for your small chicks and your eggs.
- Bafana** *What measures are needed to prevent the culling teams spreading the disease on their vehicles, on their clothes, or equipment?*
- Vaughan** The culling teams that move onto an affected area or affected farm will be resident there throughout the culling process, and won't be moving on and off the farm. Those teams will remain on the farm for the whole culling process and will only move out once they have been disinfected and sterilized.
- Bafana** *How then would you build trust between the culling teams and rural populations and, for that matter, farmers?*
- Vaughan** The only sensible way, or the logical way, that I can see to build up that trust is by education, which is why we are going out to educate the rural people. Once the rural people know exactly why we are culling and the reasons behind it, then there won't be this mistrust between the two teams.
- Bafana** *If a farm or a village suffers from an outbreak of avian flu, what kind of movement restrictions will be imposed?*
- Vaughan** Immediately the farm or village will go into a quarantine where there will be people in charge of quarantining that area or that farm. The quarantine zone falls into three zones. You have your infected zone, you have your control zone which will be under surveillance 24 hours a day, and then you have your disease-free zone. Once those restrictions are in place, full biosecurity measures will be installed in those three zones and will be controlled by qualified personnel. And all movement of people and poultry will be controlled. The biosecurity measures, meaning all vehicles in or off the farm or people in or off the farm have to be sterilized, all records kept of where people have come from originally, of where they are going from that place, and disinfectants in place around the whole farm. No movement of any other animal must be allowed in those areas of zoning.
- Bafana** *Inspector Vaughan, why are these restrictions necessary?*
- Vaughan** The H5N1 virus is highly contagious and can be carried on your feet, on your clothes, on your hands, and by infected poultry from bird to bird, so it is very necessary to contain that virus. Very strict biosecurity measures must be put in place with disinfectant, and to control the movement of people, as well as vehicles and poultry. *End of track.*

## Avian flu

### *Compensation - necessary for control*

**Cue:**

In March of 2006, Cameroon reported its first cases of the H5N1 avian flu virus. The outbreaks occurred in the north of the country, where most poultry are local birds kept by small-scale farmers; there are few commercial flocks. To prevent the disease from spreading, the government used the standard control strategy, slaughtering all domestic poultry, both healthy and diseased, within a 30 kilometre radius of the outbreak. People within this area were asked to declare all their poultry to the veterinary authorities, so that they could be slaughtered, a process known as culling. The government also paid financial compensation to the farmers for all birds that were slaughtered.

The outbreak of avian flu in Cameroon was successfully controlled, not least because of the compensation that was given to the farmers. So why is paying compensation so important in controlling an outbreak? To find out more about the reasons for offering compensation, and how the process worked in Cameroon, Martha Chindong spoke to Dr Linus Chimangha, an inspector in the Cameroonian ministry responsible for agriculture.

**IN:** "I will say compensation is one ..."  
**OUT:** "...have hope in our poultry sector."  
**DUR'N** 5'20"

**BACK ANNOUNCEMENT:** Dr Linus Chimangha of the agriculture ministry in Cameroon was speaking to Martha Chindong. The interview comes from a radio resource pack on avian flu produced by CTA.

**Transcript**

**Chimangha**

I will say compensation is one of the things involved in stamping out the disease. It involved culling, or the massive killing of all the birds where the outbreak was detected, and when these birds are killed we now go into compensation of the farmers involved. What happened is that a census of all the people within that radius who had poultry was conducted. They counted the chickens to know who has what, and then after that the teams which were mobilised destroyed all these birds in a systematic way, and then the farmers concerned were compensated, and the compensation which was put in place was 2000 francs per chicken or duck or goose, and for pigeons it ranged from 500 francs to 1000 francs.

**Chindong**

*Was there any form of registration, or were they just counted like this and compensated?*

**Chimangha**

No previous registration, except for people who maybe had commercial poultry flocks. But announcements were made for them to own up and come and declare that they had chickens, and when they declared that they had chickens the technicians went out to ensure that the chickens were actually seen, and then they counted them.

**Chindong**

*Was there any form of discrimination?*

- Chimangha** No discrimination whatsoever. If you had 10 chickens and they were counted and destroyed you were compensated. If you had 20, or 50... and again, whether you were a man or a woman or a child, everybody was treated just the same.
- Chindong** *You said that compensation, financial compensation was 2000 francs per chick. Is it not too small?*
- Chimangha** Not at all. Some people may consider that it is small for the fact that if they maybe took their chickens to the market they could have sold for 2500 francs or 3000, but there in the north chickens are generally very cheap. Then again, within this time of the outbreak, you know that there was this psychosis, this general depression, and people were scared of consuming poultry products. So at that time chickens were selling just too cheap. You could buy a big chicken for more than...if anything just between 500, 700 or maximum 1000 francs. Even eggs, they were being sold as cheap as 25 francs per egg. So this measure was even welcomed by the farmers for the fact that all their chickens had to be cleared away, and the government at least gave them something, so the farmers really appreciated the measures that were instituted. It was in no way small. And this helped to encourage, for the fact that there was compensation, it helped to encourage the people, so that if there were any sick fowls around, they would come out voluntarily and declare, they would not try to hide them and take them and sell them in the market clandestinely. So everything was well done, and I think it is thanks to these measures which were instituted that the situation was brought under control.
- Chindong** *Can that be termed the significance of compensation, or is there another significance?*
- Chimangha** I think this is the significance, the importance of it. First to bring the disease under control, to motivate the farmers so that they don't feel so depressed, and equally so that the farmers can have some little money and keep on in business when the disease is eventually brought under control.
- Chindong** *Now that reminds me, what is the ministry doing to re-launch this poultry sector, because farmers are really down?*
- Chimangha** We had a meeting in May, which was really a brainstorming with all the poultry producers in Cameroon and their representatives, and I think part of the request made by the farmers to the government was to see how they can mobilise funds and give some micro-credit, some loans to poultry farmers, particularly to those smallholders. And then to see how people who incurred no matter what loss around that time, they could be compensated. Because within that period of outbreak it was only this culling procedure which was implemented in the areas where the disease was discovered, it was only those people who were compensated. But many other people who incurred some losses because their products were not being bought, they have not been compensated, so there was an appeal made to government that such cases could also be considered. In Bafousan so many thousands of eggs, more than some 36,000 were lost. Here in Yaoundé, hatcheries, they destroyed a lot of day old chicks, a lot of layers; so most of those people they are praying that the government will do something.
- Chindong** *Any other thing that you would love us to discuss?*

**Chimangha**

Just to say that the general fear that was brought into people's minds with the outbreak of the bird flu, it is dying down, and really it should die down. People should not be afraid. They should consume poultry products as much as possible, but have them well boiled. Their chickens and their eggs, they should be well boiled. And people should be encouraged to go into poultry production, because with all the measures being put into place, even if the disease comes now I think we are better placed to fight it than we used to be before. So let us have hope in our poultry sector. *End of track.*